

Robotics Experts



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in places like Palm Beach County, Florida, it was an absolute necessity after the pandemic, as guests started arriving but employees did not! Cobots like Bella came to the rescue of restaurant owners there.

Cobots are being deployed in almost every field. How about Spot—the cobot from Boston Dynamics that scans a building under construction to avoid flaws, the smart cobot from Bloomfield Robotics that inspects vineries and notifies when the grapes are ready for plucking, the UR cobots used by NASA to inspect aircraft fuselage, or ABB's PixelPaint-equipped cobot that can apply any design you show it on a car's body?

Cobots being safer and less expensive are also being used by schools and colleges to teach students about automation. "As more and more businesses from different sectors begin to embrace automation, the use of collaborative robots will become more common," says Guruprasad. With existing and emerging use cases, ABI Research predicts that cobots will grow from 5% of the global robotics market to 29% in 2030.

Do we need cobots in India?

In a country like India where labour is available in surplus, there is a

general opinion that we do not need cobots. This mindset has even led to several robotics startups giving up their cobot plans. However, potential users have to evaluate cobots keeping the basic premise in mind—that cobots are here to work with human workers and not replace them. With lifestyle diseases cropping up, and health departments cracking down on various industries, it is essential to improve the working conditions of human workers by giving them an able helping hand that will reduce their physical strain, avoid unergonomic activities and keep them away from hazardous substances.

Guruprasad feels that, "In low-cost labour markets like India, utilisation of cobots is driven more by the ease-of-use and coexistence features of collaboration. There are multiple industrial applications which require even intermittent collaboration. For example, in machine tending, an associate is required to change various parameters of the machining centre on a trial-and-error basis on which the robot is being used for tending. This is possible on a seamless basis through cobots."

He adds that the market is expanding, with the rise in automation demand, the availability of inexpen-

sive, energy-efficient industrial cobots and the increase in R&D spending. Emerging economies have started adopting industrial cobot based solutions and, unlike earlier, when they were used by only the automotive and manufacturing sectors, now they are being picked up by other sectors like food and beverage, metals, chemical and material industries, aerospace, and electronics too.

Sougandh explains that cobots are also economically-viable in the Indian scenario. He says, "Repeated activities wear out employees, resulting in production mistakes. Unlike humans, cobots can work 24/7 while maintaining high accuracy and precision round the clock. This further leads to the reduction of manufacturing costs and the elimination of waste. Since cobots are easily programmable, employees without a programming background can easily operate a cobot. Hence, manufacturers need not spend additional sums to have an engineer on-site. In addition, cobots provide a rapid return on investment (RoI), and that too in a short span of 12 months."

A few years back, Aurolab, an Indian manufacturer of eyecare products, deployed cobots from UR for assisting human workers in the manufacture of their cataract kit. The manufacture of intraocular lenses requires high precision and accuracy, as it is pivotal to regaining eyesight after the surgery. After deploying eight UR robots to assist employees in tasks like material handling, there was a 15% increase in the product output, with over 2,000,000 lenses per year.

Indeed, the future of collaborative robots is getting better by the day, with users slowly understanding the benefits. According to a new report from global tech market advisory firm ABI Research, the cobot market is set to grow substantially over the coming decade. The market had a global valuation of \$475 million in 2020, which will expand to \$8 billion in 2030, with a projected CAGR of 32.5%. **EFY**